

4.0 Sockets Server and Clients

Learning Points

- ☐ Understand that the socket layer allows the Application layers to interface with the Transport layer and vice versa
 - ☐ Understand client-server and peer-to-peer network architectures, their characteristics, advantages and limitations
 - ☐ Understand the various application layer protocols and their uses
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Sockets

endpoint of connection between two devices on a network

- interfaces between transport and application layers, allows application to utilize services provided by the transport layer

Stream Socket / TCP - Connection, ordered, error checked

Datagram socket / UDP - fast unreliable communication

Client Server Architecture

- model where multiple client devices communicate and request services from the central server

Server - a device / program providing resources to clients over a network

- Infinite program unless error / shutdown
- accessed via IP and well known addresses

Advantages and Disadvantages of Client Server

Advantages

- centralized server resource managements (Database, backup and security)
- scalability - only need to upgrade server
- centralized security
- ease of maintenance
- Reliability and consistency of service (single source)
- Remote access support - work from anywhere

Disadvantages

- single point of failure
 - prone to overload / congestion
 - high cost of maintenance
 - complexity of security and enforcement
 - network dependence
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Peer to peer (p2p) Architecture

decentralized network model where each computer acts as both server / client

- services provided directly by each member

Features

- equal role
- Decentralized
- Resource sharing
- Autonomy

Advantages

- low cost
- scalable
- no single point of failure
- resource sharing -files, processing and bandwidth

Disadvantages

- lacks central control
 - security risk - malware spread as there is no centralized security
 - poor data consistency
 - low performance with peers that have low bandwidth
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Application protocols:

Dynamic host configuration protocol - automatic assignation of IP address and configurations over the network

- IP, subnet mask, Default gateway, DNS server address, Lease Duration...ect

Follows DORA for assignation:

1. Discover - client performs broadcast to find DHCP servers
2. Offer - server replies with offer including IP and its configurations
3. Request - client requests to use the offered IP

4. Accept - Server confirms and leases the IP

* DHCP uses UDP and Broadcast

- UDP has minimal overhead
- requesting client does not have actual IP to use for DHCP server

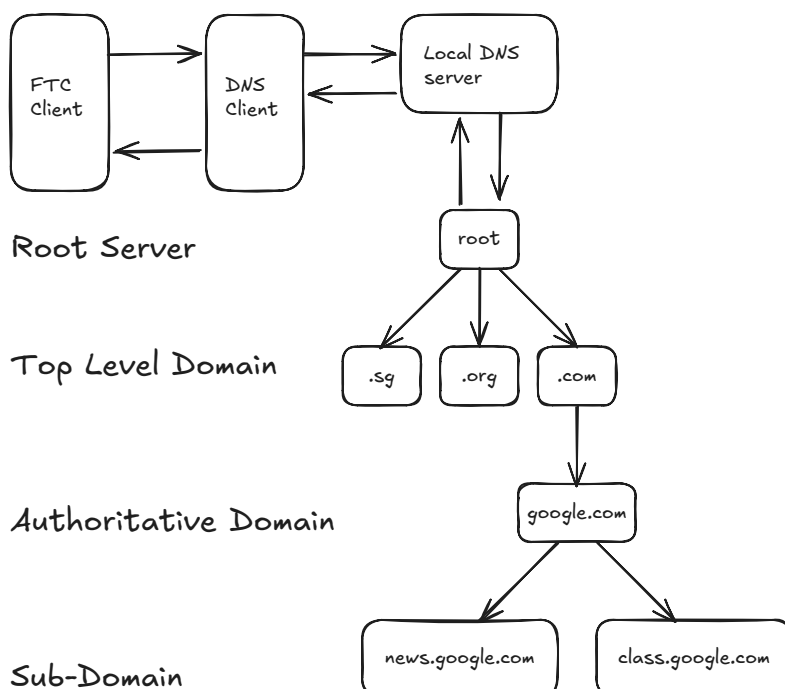
Types of allocation:

- static:
 - Manual, self assigned
 - requires reservation and always returns same IP from DHCP
- Dynamic:
 - Server leases reusable IP to client based on Lease Duration

Domain Name System - translates domain names into usable IP address for routing

Translation process:

1. User passes domain name to File transfer Client (FTC)
 2. FTC passes DNS client File Transfer server name
 3. DNS client sends a message to the DNS server containing File transfer server name, via the IP of the DNS server
 4. DNS server responds with IP of File Transfer Server
 5. DNS client passes IP to File Transfer client
 6. Client accesses Server via IP
- * DNS client and local DNS server both check cache before passing request on, read below



Hierarchy

Order of which servers are accessed during DNS

Tier	Example
Root	top server for all domains
Top Level Domain	.net, .edu, .gov ...ect
Authoritative Domain	google.net ...ect
Subdomain	news.google.com ...ect

DNS resolution

1. Check local cache
 2. Query Local DNS server and its cache
 3. server Queries Global DNS
- Recursive resolution - local server queries the root server, where each server along the hierarchy of name servers queries the next server on the hierarchy until a logical address is found.
 - Iterative - local server manually queries each server on the hierarchy and obtains address of next server until logical address is found

Hypertext Transfer Protocol (HTTPS) - allows clients to fetch hypertext documents, images and scripts from a web server

- use port 80 and 443, operates off client-server model
- stateless - no information is collected by server on client

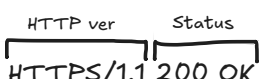
Request:

Request Line: 

Header Lines: some json esque format,
additional info to server

Message Body: Optional, for POST or submitted Data ...ect

*Response:

Request Line: 

Header Lines: some json esque format,
additional info to server

Message Body: Optional, for POST or submitted Data ...ect

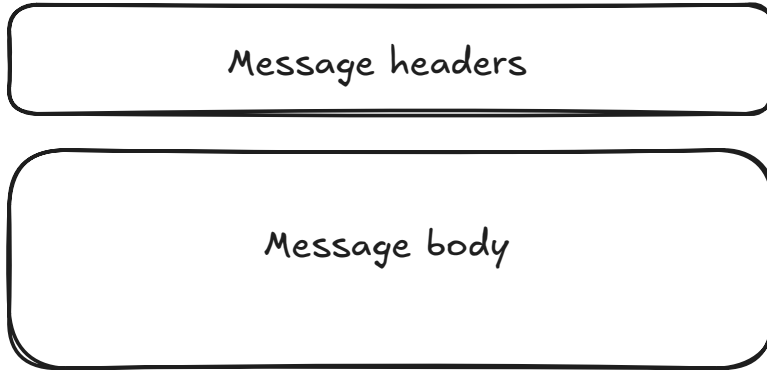
Simple Mail Transfer protocol (SMTP) - sends / relays mail across the internet

* NOT INVOLVED IN RETRIEVAL OF MAIL

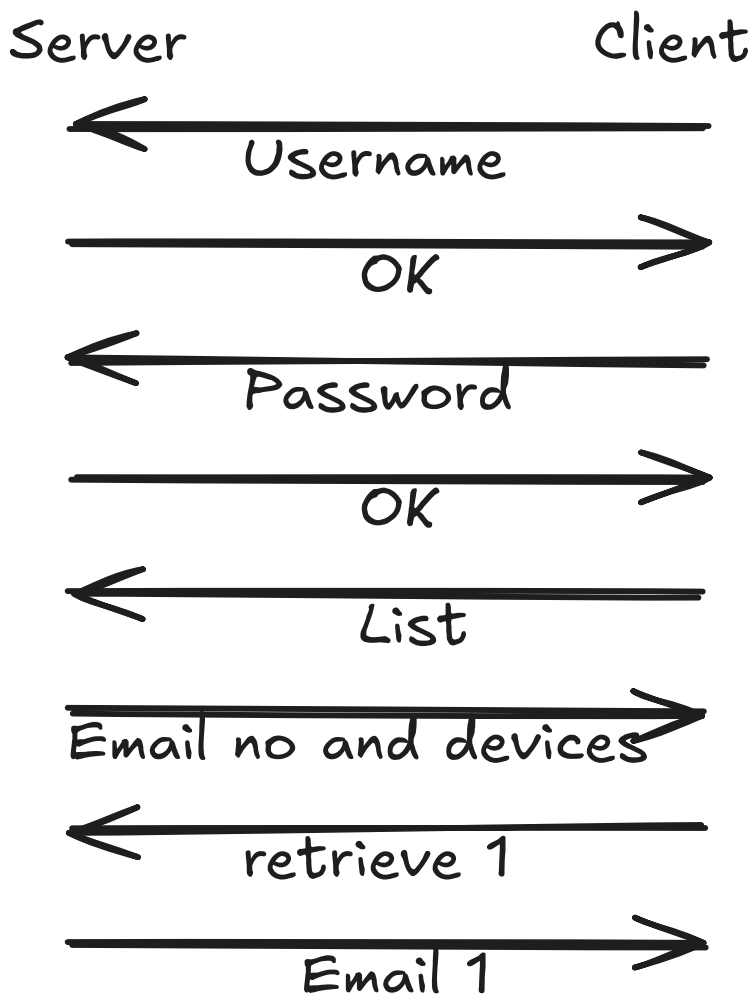
- Uses port 25 and 578
- Operates off client server model

Message:

Envelope (Control metadata): mail from
recp to
Data



Post Office Protocol 3 - allows for the donloading of email from server to client and deletion of messages on the server



Internet Message Access Protocol V4 - allows user to view and manage messages directly from server

- uses ports 143 and 993
- user can access from any location and ensures all copies are synced and consistent

File Transfer Protocol (FTP) - protocol for copying a file from one host to another

- uses port 21 for control information and port 20 for data transfer
- automatically resolves the following
 - different naming conventions
 - different ways of data representation
 - different directory structures